WORKED SOLUTION

STEP I 2012, Question 9

Mechanics • Projectile motion

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Read the question before you go on. It is in the STEP Support Programme's question database:

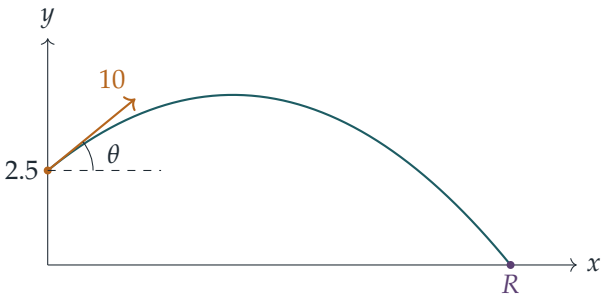
step.maths.org/questions/12-s1-q9

The key idea

This is a projectile launched from a height, so the time of flight comes from a quadratic in t, and the question is really about rewriting its root in terms of $c = \cos 2\theta$. The double-angle formulae $\sin^2\theta = \frac{1}{2}(1-c)$ and $\cos^2\theta = \frac{1}{2}(1+c)$ do all the work. Once the range is a function of the single variable c, maximising it is ordinary calculus, with a little care over squaring.

Solution

We model the shot as a particle moving freely under gravity, with no air resistance. Take the origin on the ground directly below the point of projection, with x horizontal and y vertically up, and assume $0 \le \theta < 90^\circ$ (the shot is thrown forwards and upwards), so that $\sin\theta \ge 0$ and $\cos\theta > 0$.



The time of flight

The horizontal velocity stays at $10\cos\theta$, and the vertical motion has initial velocity $10\sin\theta$ and acceleration -10, so at time t

$$x = 10t\cos\theta, \qquad y = 2.5 + 10t\sin\theta - 5t^2.$$

The shot lands when $y = 0$, that is, when

$$5t^2 - 10t\sin\theta - 2.5 = 0 \quad \Longleftrightarrow \quad t^2 - 2t\sin\theta - \tfrac{1}{2} = 0.$$

The product of the roots is $-\frac{1}{2} < 0$, so there is exactly one positive root, and it is the one with the plus sign:

$$T = \sin\theta + \sqrt{\sin^2\theta + \tfrac{1}{2}}.$$

Now use $\sin^2\theta = \frac{1}{2}(1 - \cos 2\theta) = \frac{1}{2}(1 - c)$. Since $\sin\theta \geq 0$ we may write $\sin\theta = \sqrt{\frac{1}{2}(1-c)}$, and $\sin^2\theta + \frac{1}{2} = \frac{1}{2}(2 - c)$. Hence

$$T = \sqrt{\frac{1-c}{2}} + \sqrt{\frac{2-c}{2}} = \frac{1}{\sqrt{2}}\left(\sqrt{1-c} + \sqrt{2-c}\right),$$

which is the required time in seconds.

The range

Similarly $\cos^2\theta = \frac{1}{2}(1 + c)$, and $\cos\theta > 0$, so $\cos\theta = \sqrt{\frac{1}{2}(1+c)}$. The range is

$$\begin{aligned} R = 10T\cos\theta &= 10\sqrt{\frac{1+c}{2}} \cdot \frac{1}{\sqrt{2}}\left(\sqrt{1-c} + \sqrt{2-c}\right) \\ &= 5\left(\sqrt{1-c^2} + \sqrt{(1+c)(2-c)}\right) = 5\left(\sqrt{1-c^2} + \sqrt{2+c-c^2}\right) \text{ m}. \end{aligned}$$

As θ runs over $[0°, 90°)$, $c = \cos 2\theta$ runs over $(-1, 1]$, so we maximise R over this interval.

Maximising the range

For $-1 < c < 1$, differentiating,

$$\frac{\mathrm{d}R}{\mathrm{d}c} = 5\left(\frac{-c}{\sqrt{1-c^2}} + \frac{1-2c}{2\sqrt{2+c-c^2}}\right).$$

A stationary point needs

$$\frac{c}{\sqrt{1-c^2}} = \frac{1-2c}{2\sqrt{2+c-c^2}}. \qquad (*)$$

Ask yourself Can we just square $(*)$, and what might go wrong?

Squaring can introduce false roots, because it loses the signs. Both denominators are positive, so in $(*)$ the numbers c and $1 - 2c$ must have the same sign (or both be zero, which is impossible). That forces $0 < c < \frac{1}{2}$, and we keep this in mind. Squaring and

clearing denominators:

$$\begin{aligned} 4c^2(2+c-c^2) &= (1-2c)^2(1-c^2) \\ 8c^2+4c^3-4c^4 &= 1-4c+3c^2+4c^3-4c^4 \\ 5c^2+4c-1 &= 0 \\ (5c-1)(c+1) &= 0. \end{aligned}$$

The root $c=-1$ is outside $0<c<\frac{1}{2}$ (it is the vertical throw), so the only stationary point is $c=\frac{1}{5}$. It really does satisfy $(*)$: both sides equal $\frac{1}{2\sqrt{6}}$, since $\sqrt{1-\frac{1}{25}}=\frac{2\sqrt{6}}{5}$ and $\sqrt{2+\frac{1}{5}-\frac{1}{25}}=\frac{3\sqrt{6}}{5}$.

At this point

$$R(\tfrac{1}{5}) = 5\left(\frac{2\sqrt{6}}{5}+\frac{3\sqrt{6}}{5}\right) = 5\sqrt{6} \approx 12.25.$$

To see that this is the greatest value, compare with the ends of the interval: $R\to 0$ as $c\to -1$, and $R(1)=5\sqrt{2}\approx 7.07$. Since R is continuous on $[-1,1]$ and differentiable inside, its maximum is at an end or at a stationary point; the only stationary point gives the largest of these values. So the range is greatest when $c=\frac{1}{5}$, as required. (This is $\theta=\frac{1}{2}\arccos\frac{1}{5}\approx 39.2^\circ$, a little below 45°.)

The extra distance

At 45° we have $c=\cos 90^\circ = 0$, so

$$R(0) = 5\left(1+\sqrt{2}\right).$$

The extra distance is therefore

$$5\sqrt{6}-5\left(1+\sqrt{2}\right) = 5\left(\sqrt{6}-\sqrt{2}-1\right)\ \text{m},$$

as required.

Checks

- At $\theta=0$ ($c=1$) the time is $\frac{1}{\sqrt{2}}$, matching $2.5=5t^2$ for a ball rolled off a 2.5 m ledge.
- As $c\to -1$ (vertical throw) the range tends to 0, as it should.
- Numerically, the extra distance is $5(2.4495-1.4142-1)\approx 0.176$ m, only about 18 cm: the optimum is very flat. A step-by-step simulation of the flight, and a fine grid search of $R(c)$, both agree with the formulas and put the maximum at $c=0.2$.

Common slips

- Taking the minus sign in the quadratic formula, which gives a negative time.

- Squaring $(*)$ and keeping $c = -1$ without checking it, or forgetting to justify that $c = \frac{1}{5}$ is a maximum rather than a minimum.
- Assuming the best angle is 45°: that is only true when launch and landing are at the same height.

Beyond STEP

Launching from a height always lowers the optimal angle below 45°; in general the best angle satisfies $\tan\theta = u/\sqrt{u^2 + 2gh}$, which here gives $\tan\theta = \sqrt{2/3}$, the same as $\cos 2\theta = \frac{1}{5}$. Optimising a function of a parameter, and noticing how flat the maximum is, is a first taste of sensitivity analysis: near a maximum, a small error in the parameter costs only a second-order loss. For more on projectiles with and without air resistance, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

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